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Synthesis of tantalum oxide thin films using RF magnetron sputtering for RRAM application

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ABSTRACT

Resistive Random Access Memory (RRAM) is a prominent contender in the realm of emerging non-volatile memory technologies owing to its numerous advantages like simple structure, low power usage, high speed, and exceptional scalability. In the current study, RRAM device was fabricated using tantalum oxide (TaO_x) as a resistive switching insulating oxide layer, which was sandwiched between two conducting electrodes, namely Ag and ITO. RF magnetron sputtering was used to deposit the insulating layer of TaO_x at room temperature on the ITO substrate. Further, X-ray reflectivity and hard X-ray photoelectron spectroscopy analyses were conducted to find the thickness, roughness, and elemental composition of the deposited TaO_x films. Using a semiconductor analyzer, the RRAM cell's first DC switching characteristics were determined. The data acquired exhibited a resistance ratio (R_{HRS}/R_{LRS}) of 120 showing a good distinction between the two-resistance states. Additionally, a thorough investigation was conducted to examine the electrical conduction mechanism occurring inside the fabricated RRAM device.

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KEYWORDS
Metal oxide; resistive switching; RF sputtering; SCLC mechanism

1. Introduction

In today's digital era, there is an insatiable need for memory devices that offer improved density, enhanced performance, reduced cost, and faster speeds. Contemporary memory systems such as flash memory, static random-access memory (SRAM), and dynamic random-access memory (DRAM) are encountering difficulties such as slow data transfer rates, excessive power usage, and limitations in terms of scalability. All these contemporary memory devices work on the principle of charge storage. SRAM stores charge at the nodes of cross-coupled inverters, while DRAM stores data in the form of charge at the capacitor. Flash memory, on the other hand, saves information by storing charge at the floating gate of transistors. These existing memory technologies are facing challenges in terms of downsizing. This is mainly because of the degradation of their performance

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caused by the loss of charge at the nanoscale (*1*). Consequently, researchers are seeking an alternative memory technology with the capacity to supplant these existing memory technologies. The principles of change in resistance underpin the operation of these new emerging memory technologies, which include phase change random access memory (PCRAM), spin-transfer torque magnetoresistive random-access memory (STT-MRAM), and resistive random-access memory (RRAM). Among these, RRAM is one of the standout non-volatile emerging memory (NVeM) technologies because of the several advantages it offers, including simplicity of structure, reduced consumption of power, integrability with existing complementary metal oxide semiconductor (CMOS) technology, higher endurance, and excellent scalability (*2*–*4*). Radiation-hardened memory devices, non-volatile logic functions, hardware-based security primitives, neuromorphic computing, and multi-level cell storage are just a few application areas of RRAM devices (*5*–*7*). The sandwich configuration of RRAM is a metal-insulator-metal (MIM) arrangement, which puts an insulator layer between two metal electrodes. RRAM works on the principle of resistive switching (RS). In response to an applied external voltage above a certain value, the insulating layer often called the RS layer, goes through a resistance state transition, going from a state of high resistance (HRS) to a state of low resistance (LRS). Data or information is stored in RRAM devices by use of these resistance states. The HRS (off state) and LRS (on state) denote the logical "0" and "1" in the binary form. As the resistance states can hold the data even after the removal of voltage supply, it shows the non-volatile characteristics of RRAM devices (*8*).

Several materials including binary oxides, perovskite oxides, chalcogenides, ferrites, organic materials like biomaterials and polymers, etc. have been used as RS layer in RRAM devices. Binary transition metal oxides like TiO_x, HfO_x, ZrO_x, ZnO, NiO, SiO_x and Tantalum oxide (TaO_x) are promising options for the RS layer because of their exceptional scalability, simple fabrication process, high-density integration, significant resistance ratio, low cost, and high speed (*10*–*13*). The TaO_x-based RRAM devices have gained considerable attention due to their high dielectric constant ($k > 25$), high thermal stability, high cyclic endurance (10^{12} cycles), fast switching speed (100 ps), ease of deposition using conventional techniques, chemical inertness, and CMOS compatibility (*10*, *11*, *9*). In recent studies, researchers have explained the RS mechanism by a combination of the space charge limited currents (SCLC) effect and oxygen vacancies (V_o) or defects, which naturally exist in the oxide layer in RRAM devices. Oxygen non-stoichiometry and oxygen-deficient oxide phases are the most fundamental factors leading to reversible RS in TaO_x films (*12*, *13*). The regions of higher V_o concentration facilitate the formation of conductive filament within the insulating layer. Conductive nano filament formation or rupture corresponding to connecting or disconnecting the two metal electrodes can cause the device to switch between LRS and HRS, which explains the sharp set and reset process in the RRAM devices.

More recently, Yun et al. (*14*) fabricated a Pt/TaO_x/TiN structure RRAM device and evaluated the device's volatile and nonvolatile properties through various compliance current experiments. It was observed that SCLC dominated the high resistance state in the reset mechanism. Similarly, Ding et al. (*15*) deposited TaO_x thin film to develop a Ta/TaO_x/ITO structure to show RS behavior, and the conduction mechanism of this structure is intricately linked to the concentration of oxygen ions, which was governed by the partial pressure of oxygen gas during deposition at the interface between the dielectric layer and the electrode.

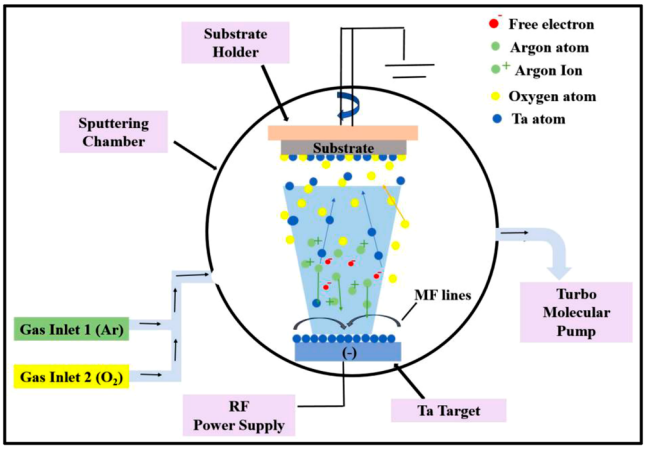


Figure 1. Schematic of RF magnetron sputtering technique.

In the present work, we fabricated an Ag/TaO_x/ITO RRAM structure, which exhibited a good ON/OFF ratio of 120 and a sharp set voltage at 0.87 V. The device showed an Ohmic conduction mechanism in both the HRS and LRS during the RESET process. For the SET process, the SCLC mechanism was used to elucidate the HRS and an Ohmic mechanism in the LRS.

2. Experimental details

A 14.7 nm thick TaO_x thin film was deposited onto a substrate consisting of silicon (Si (100), p-type) and indium tin oxide (ITO) coated glass. The glass substrate with ITO coating was directly bought from Shilpa Enterprises, India, of the product series ITO10X. The ITO layer on the glass substrate has an approximate thickness in the range of 180 ± 15 nm and a sheet resistivity value of ~ 10 ohms/sq. Prior to deposition, ITO and Si substrates were cleaned using acetone and deionized water. The TaO_x film was deposited on the ITO substrate, where a portion of it was shadow masked with aluminum foil. TaO_x thin films were deposited at room temperature (RT) using the RF magnetron sputtering technique. For deposition a 5-mm thick, 2-inch tantalum (Ta) target (purity 99.99%) was used, and a 100 W RF power supply was applied to the target. Sputtering pressure was kept at 6×10^{-3} mbar while oxygen and argon gas were introduced into the chamber with a flow rate of 18 and 12 sccm, respectively. The base pressure of 4.5×10^{-6} mbar was maintained before deposition. The Ta target was pre-sputtered for 5 min before deposition for more pure film deposition. To deposit a consistent or uniformly thick TaO_x layer, the distance between the target and the substrate was maintained at 15 cm (about 5.91 in) and the substrate rotation speed was sustained at 8 rpm during the deposition process. The RF magnetron sputtering system used here was bought from Excel Instruments. The instrumentation has been designed for various thin-film deposition of metals, alloys, compounds, and oxides (*16*). A schematic of the RF magnetron sputtering technique used in this work is represented in Figure 1.

Further, X-ray reflectometry (XRR) measurements were performed using the Malvern Panalytical (Empyrean series) apparatus using Cu kα radiation ($\lambda = 1.54$ Å) to estimate the

thickness of deposited thin film on Si substrate. Hard X-ray photoelectron spectroscopy (HAXPES) measurements were carried out at the photoelectron spectroscopy beamline of the Indus-II synchrotron facility at RRCAT, Indore. The HAXPES spectra were recorded using 15 keV Phoibos 225 H V hemispherical analyzer with incident X-ray energy of 4365 eV. The B1500 semiconductor analyzer was linked to a probe station, which allowed for electrical characterizations such as I-V measurements. Silver (Ag) paste acquired from RS Pro (silver loaded epoxy adhesive 186-3616) mixed with thinner liquid was used as the top metal electrode on the TaO_x film for conducting electrical measurements.

3. Result and discussion

3.1. X-ray reflectivity

The XRR pattern plotted against X-ray exchanged momentum (q) and XRR intensity is shown in Figure 2. The thin film deposited on a silicon substrate was used for XRR characterization. The thickness of the active layer plays a crucial role in the switching mechanism of the memory device, particularly during the process of development and rupture of conductive filament within the oxide layer. Parratt's formulism (*17*), was used to fit the XRR pattern, taking the film thickness and roughness into account as fitting parameters. In this case, the red curve corresponds to the most accurate fit to the experimental data, while the blue curve represents the experimental data. The XRR fitting indicated a thickness of 14.7 nm and a roughness value of 1.5 nm for the deposited TaO_x film on the Si substrate. The value of q was calculated using the formula

$$q = \frac{4\pi}{\lambda} \sin\left(\frac{2\theta}{2}\right) \tag{1}$$

where, λ is the wavelength of incoming X-ray, and 2θ represents the angle between transmitted and reflected beams.

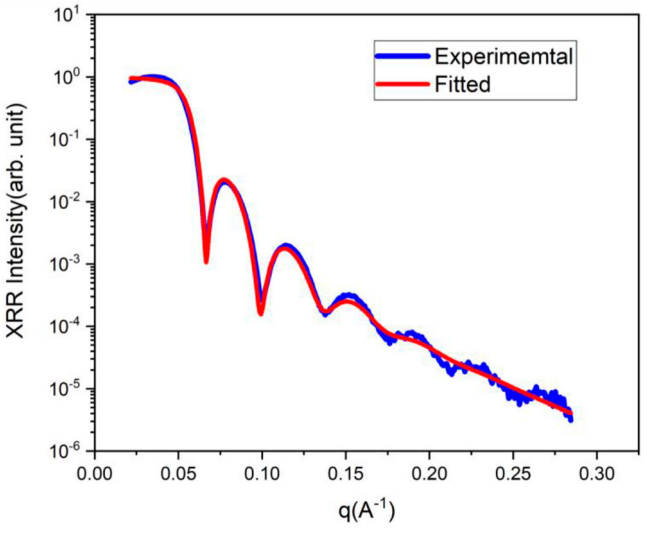


Figure 2. XRR pattern for deposited tantalum oxide film on Si substrate.

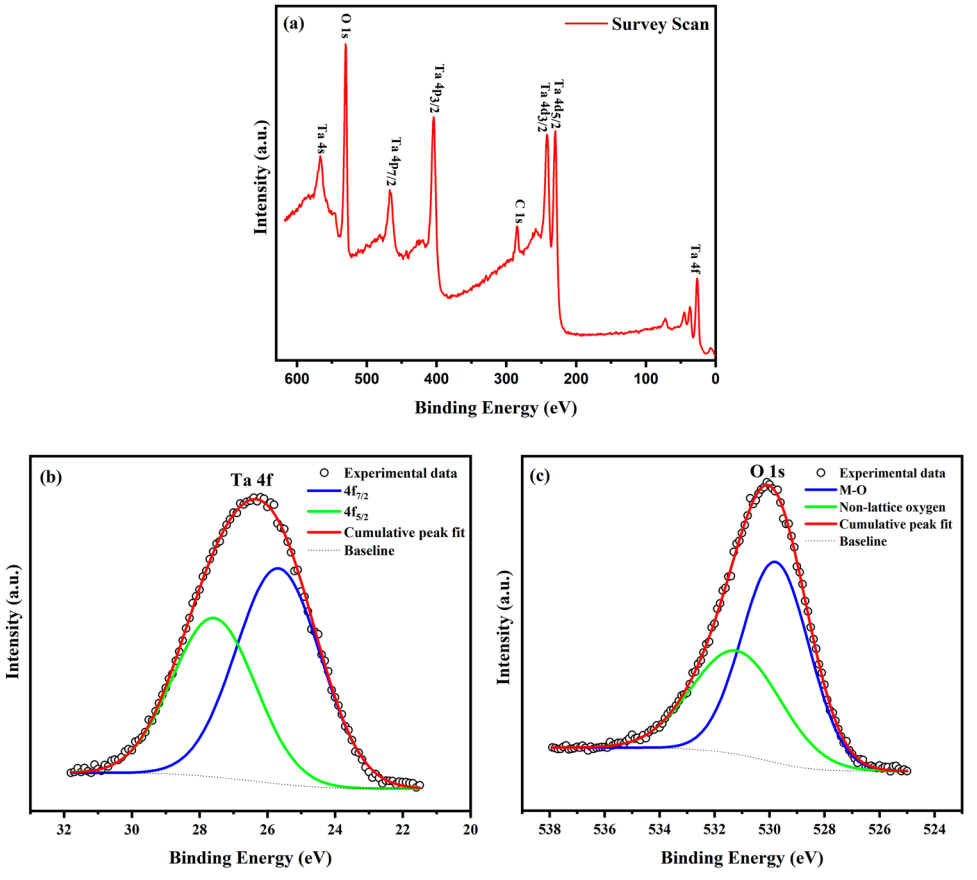


Figure 3. (a) Survey spectrum for TaO_x thin films. Deconvoluted peaks of (b) core Ta 4f spectra, and (c) core O 1s spectra.

3.2. Hard X-ray photoelectron spectroscopy

To identify the elemental composition of the deposited TaO_x thin films HAXPES characterization technique was used. Figure 3 displays the survey spectrum of the sample, which clearly indicates the presence of Ta and O elements in the film. Additionally, core level spectra for Ta 4f and O 1s are represented in Figure 3(b,c) respectively. The standard position of the C 1s peak, having BE of 284.7 eV, is used to do carbon correction for any charge induced peak shifts. The multiple peaks in the spectra were fitted using the Gaussian function following Shirley type background subtraction. The deconvoluted spectra for Ta 4f show two peaks of Ta $4f_{5/2}$ and Ta $4f_{7/2}$ located at 27.57 eV and at 25.67 eV respectively attributed to the Ta-O bonds. The peak binding energies centered at 529.7 eV and 531.2 eV for O 1 s orbital corresponds to lattice oxygen (Ta-O) and non-lattice oxygen, respectively (*18*) in the sputtered films. Further, we elucidated the elemental composition of the active layer by integrating the peak area and employing the associated relative sensitivity factor (RSF) (*19*) calculated for each element. The resulting RSF values are 3.02 for O 1 s, 1.86 for Ta $4f_{5/2}$, and 2.29 for Ta $4f_{7/2}$. The calculated atomic ratio of oxygen to tantalum (O/Ta) is 1.84,

resulting in an elemental composition of $TaO_{1.84}$. This XPS study reveals the formation of sub-stoichiometric tantalum oxide thin films.

3.3. Electrical characterizations

3.3.1. I-V characterization

To study the switching behavior, I-V curve characteristics were carried out using a B1500 semiconductor analyzer at room temperature. While performing electrical measurements, the ITO acted as the bottom metal electrode, and Silver (Ag) paste was used to make the top metal electrode. The fabricated Ag/TaO_x/ITO RRAM cell structure is shown in Figure 4 along with the I-V measurement setup.

The application of a bias voltage sequence consisting of 0 V, $+V_{max}$, 0 V, $-V_{max}$, and 0 V is generally used to study the fundamental switching mechanism in RRAM device (*20*). The transition from a HRS to LRS is referred to as SET process, and the voltage at which it occurs is called set voltage (V_{SET}). Moreover, on the application of a negative bias voltage, the device transitions from a LRS to a HRS, representing a reset process and the voltage at which it occurs is the reset voltage (V_{RESET}). In general, RS devices exhibit two types of RS modes: unipolar resistive switching (URS) and bipolar resistive switching (BRS). In the URS device, the switching between set and reset states depends on the amplitudes of voltages rather than the polarity of the voltage that is applied, although it can be altered by controlling the current compliance which is utilized in the set process and left free on the same polarity during the reset procedure (*1*,*21*,*22*). In BRS mode the switching direction is determined by the polarity of voltage, if the set process is taking place for a positive voltage, then the reset process will take place for a negative voltage. The term "resistance ratio" refers to the ratio between the two distinct resistance states, that is HRS and the LRS (*23*). The significant resistance ratio ensures a clear distinction between the two memory states, hence reducing the probability of errors during read and write operations.

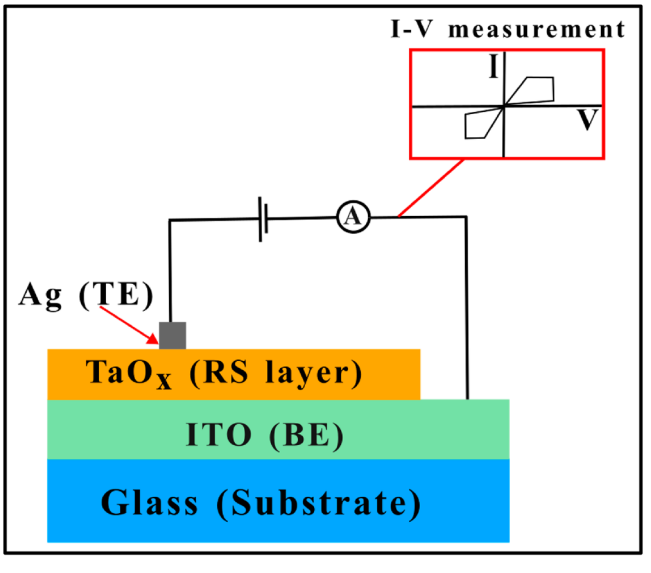


Figure 4. Schematic of RRAM device along with I-V measurement setup.

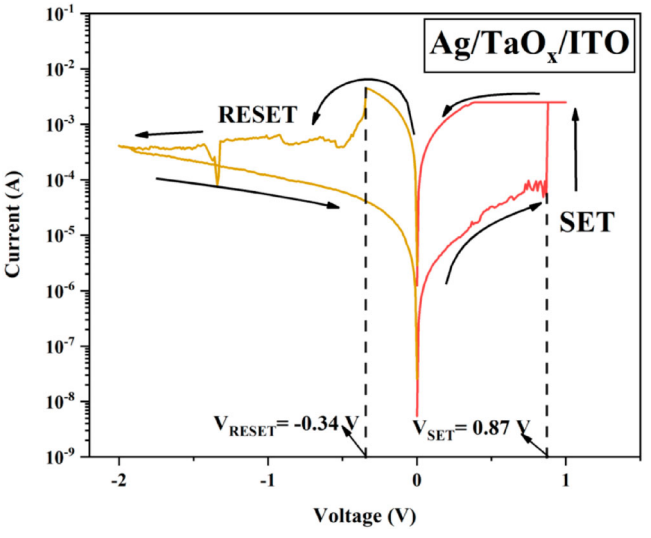


Figure 5. Standard I-V characteristics of Ag/TaO$_x$/ITO device measured at room temperature (the voltage sweep direction is shown by the arrows).

A voltage bias between the Ag (top) and ITO (bottom) metal electrodes was applied, ranging from +2 V max to −2 V max. The as-deposited sample (pristine) is usually in HRS or OFF state. Applying a positive bias voltage causes the device to switch from off-state to on-state, *i.e.* HRS to LRS at a set voltage value of 0.87 V. Thus, we observed a sharp resistance switching from HRS to LRS in the set process. On applying the negative polarity of voltage, the device started a reset process. However, the reset process was gradual in nature, the current decreases gradually from −0.34 V to −1.98 V. Typical BRS behavior is exhibited by the fabricated Ag/TaO$_x$/ITO RRAM device, where the set process happens for positive polarity of voltage and the reset process occurs for negative polarity of voltage. The I-V characteristics of the memory device are displayed in Figure 5, and the resistance vs time curve is shown in Figure 6. An observed resistance ratio or memory window (R_{HRS}/R_{LRS}) of around 120 was recorded for the device and both the HRS and LRS show stable characteristics over 1.2×10^3 s which was read at 0.2 V, clearly delineating the differentiation between the two states of resistance.

3.3.2. Conduction mechanism

There are several types of electrical conduction mechanisms like Ohmic, SCLC, tunneling, hopping, standard Schottky, and pool Frenkel conduction to explain the behavior of RRAM devices (*24*). These mechanisms liable for RS behavior are reliant upon the characteristics of conductive filaments, such as metal ions or oxygen vacancies, which enable conduction. In these different kinds of conduction mechanisms (*23*, *25*, *26*), the relationship between the current and voltage is described as follows:

For Ohmic conduction: $I \propto V$

For Pool-Frenkel emission: $\log (I/V) \propto V^{1/2}$

For Schottky emission: $\log (I) \propto V^{1/2}$

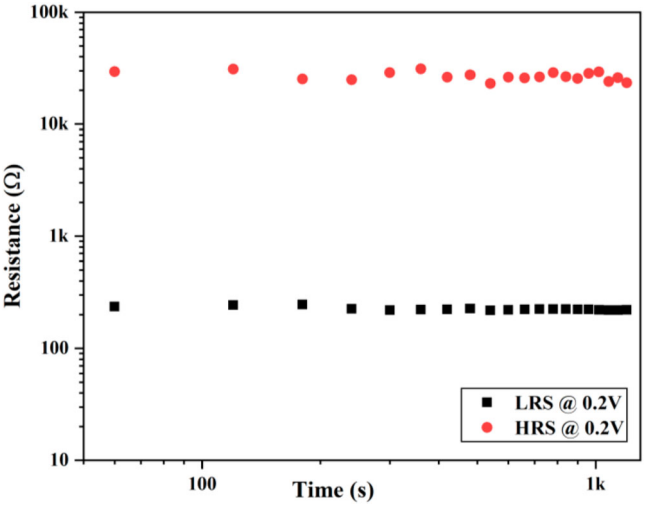


Figure 6. Resistance vs time curve at 0.2 V read voltage for the Ag/TaO_x/ITO device.

In the context of SCLC three separate regions to consider: (a) Ohmic region where $I \propto V$ (b) Child's square law region where $\propto V^2$ and, (c) the sharp current increase region.

The mathematical equation describing the correlation between log (I) and log (V) in the context of Ohm's law is represented as (*27*, *28*):

$$\log(I) = \log(V) + \log\left(\frac{en_0mS}{d}\right) \tag{2}$$

where e represents the elementary charge, n_0 denotes the density of the thermally generated free carrier, m is the carrier mobility, S is the area of the top metal electrode, and d denotes the thin film thickness.

Similarly, the log(I) and log(V) relationship in the context of the SCLC mechanism is written as follows:

$$\log(I) = 2\log(V) + \log\left(\frac{9\varepsilon_0\varepsilon_r m}{8d^3}\right) + \log(S) \tag{3}$$

where ϵ_0 represents the permittivity of free space and ϵ_r denotes the static dielectric constant.

By plotting the ln (I) vs ln (V) curve, we can conclude that the device obeys Ohm's law when the slope of ln (I) to ln (V) is 1, if the value of slope is 2 then the mechanism is explained by SCLC and further if the slope value is > 2 then the conduction behavior is explained by trap filled SCLC (*27*, *29*).

So, to better understand the dominant electrical conduction mechanism of the fabricated TaO_x based RRAM device, we plotted double (ln-ln) I-V curves for both the set and reset processes. The achieved results are shown in Figure 7. Most of the adjusted R-square values for the linear fitting curves fall within the range of 0.94 to 0.99. Figure 7 (a) along with (b) displays the double log curves and the fitting data for the Ag/TaO_x/ITO device's set and reset state, respectively. In this study, we found that the LRS exhibits Ohmic conduction behavior for both positive and negative regions, *i.e.* for set and reset processes, with a slope value of ~ 1.04 (adj. R^2 is 0.99) and ~ 1 (adj. R^2 is 0.99) respectively. However, the HRS

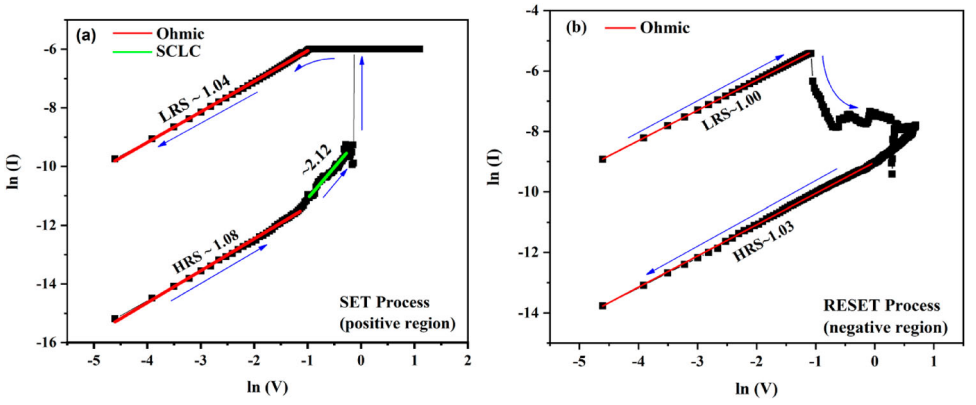


Figure 7. Analysis of different conduction mechanisms for both (a) set process and (b) reset process.

Table 1. Comparison between various single layer TaOx based RRAM devices.

S. No.	Device structure	Deposition method for RS layer	RS layer thickness (nm)	V_{RESET} (V)	V_{SET}(V)	RSmode	Resistance ratio	References
1.	Ag/TaO_x/TiN	DC reactive sputtering	~ 15	−1	1	Bipolar	~ 73.2	(*30*)
2.	ITO/TaO_x/ITO	DC sputtering	~ 25	−4.0	3	Bipolar	~ 100	(*31*)
3.	Ta/Ta_2O_5/Pt	Electrochemical synthesis	~ 20	−1.9	1.25	Bipolar	> 10	(*32*)
4.	Ta/TaO_x/SS	Magnetron sputtering	~ 20	−0.6	1.5	Bipolar	~ 10^3	(*33*)
5.	TiN/Ta_2O_5/Pt	RF sputtering	~ 30	−1.5	1.5	Bipolar	~ 10	(*34*)
6.	Ag/TaO_x/ITO	RF magnetron sputtering	~ 15	−0.34	0.87	Bipolar	~ 1.2 × 10^2	Present work

in the set process shows a non-ohmic conduction mechanism. We can see that in the low voltage region (0 V–0.38 V) the slope value is ~ 1.08 (adj. R^2 is 0.99) and then it increases to ~ 2.12 (adj. R^2 is 0.94) in the higher voltage region (0.39 V–0.86 V), and the sharp increase in the current at the V_{set} value (0.87 V) indicating an Ohmic to SCLC conduction mechanism. A slope value of 1.03 (adj. R^2 is 0.99) revealed that HRS exhibited Ohmic conduction behavior during the reset procedure.

The Ohmic conduction mechanism is attributed to the small number of mobile charge carriers produced as a result of thermal excitation. On the other hand, for SCLC, the injected charge carrier (n_i) dominates the thermally generated charge free carrier (n_o) in the conduction mechanism (*25*). The formation of a conductive filament inside the insulating TaO_x layer is indicated by a sharp increase in current at SET voltage.

Table 1 shows a comparison between previous studies carried out using a single layer of TaO_x-based RRAM and the results obtained in this work in terms of V_{SET}, V_{RESET}, RS mode, and resistance ratio. Based on the achieved results, this work presents a low-thickness, stable bipolar RRAM device using RF magnetron sputtering which is able to switch at lower switching voltages (less than 1 V) and exhibiting a reasonable memory window.

4. Conclusions

In this work, we have fabricated an Ag/TaO_x/ITO RRAM device with stable bipolar resistive switching properties using the versatile and efficient RF magnetron sputtering technique. The deposited TaO_x thin film was found to have a thickness of ~ 14.7 nm and an elemental

composition of $TaO_{1.84}$. The device displayed consistent stable bipolar resistive switching characteristics with a promising resistance ratio of 120. Further, the RS mechanism was investigated, and the conduction mechanism is discussed in the light of a linear fitting of ln (I)- ln (V) curve. The LRS for both set and reset processes demonstrated an Ohmic conduction behavior, whereas the HRS in the set process demonstrated the SCLC conduction mechanism and the HRS in the reset process demonstrated an Ohmic conduction.

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